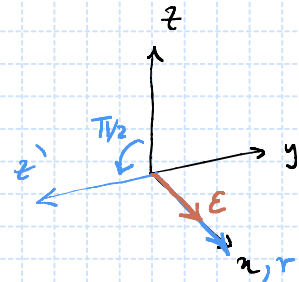
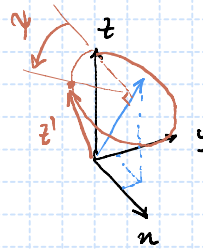
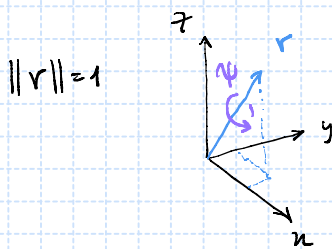


① (a) $Q = \eta + i \varepsilon_x + j \varepsilon_y + k \varepsilon_z = \begin{bmatrix} \eta, \varepsilon \end{bmatrix} = \begin{bmatrix} \eta, \begin{pmatrix} \varepsilon_x \\ \varepsilon_y \\ \varepsilon_z \end{pmatrix} \end{bmatrix} = \begin{pmatrix} \eta \\ -\varepsilon_x \\ \varepsilon_y \\ \varepsilon_z \end{pmatrix}$

$\uparrow$ scalar
 $\uparrow$ vector

(b) vector - angle

example:



$$Q = \begin{bmatrix} \eta, \varepsilon \end{bmatrix} = \begin{bmatrix} \cos(\frac{\psi}{2}), r \sin(\frac{\psi}{2}) \end{bmatrix}$$

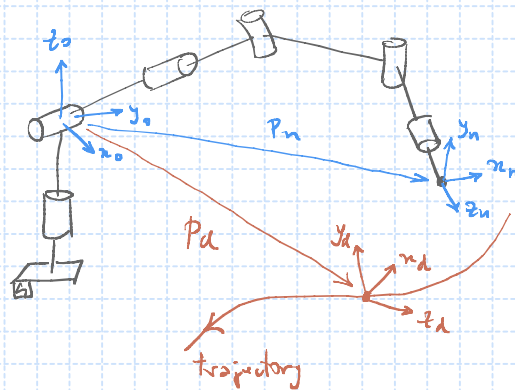
$$Q = \begin{bmatrix} \cos(\frac{\pi}{4}), \sin(\frac{\pi}{4}) \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \end{bmatrix} \\ = \begin{bmatrix} \frac{\sqrt{2}}{2}, (\frac{\sqrt{2}}{2} \ 0 \ 0)^T \end{bmatrix}$$

(c) In the vector - angle representation. If $\psi=0$ (null rotation), the direction is arbitrary, with $\|r\|=1$.

In the quaternion representation there is no singularity, since

$$Q(\psi=0, r) = \begin{bmatrix} \cos(0), r \sin(0) \end{bmatrix} = \begin{bmatrix} 1, (0 \ 0 \ 0)^T \end{bmatrix}$$

②



actual position
and orientation

desired position
and orientation

trajectory

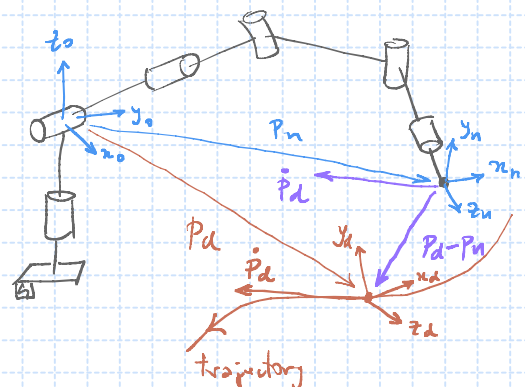
• Control action for the position,

$$\dot{P}_n = M_p \dot{q}$$

Attribute a velocity, meaning to $P_d - P_n$,

$$\dot{P}_d + K_p (P_d - P_n) = M_p \dot{q}$$

position
error (error vector)



$$\dot{q} = \mathcal{Y}_p^{-1} [\dot{p}_d + K_p(p_d - p_n)]$$

K_p is positive definite. Example: $K_p = k_p \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$, $k_p > 0$.

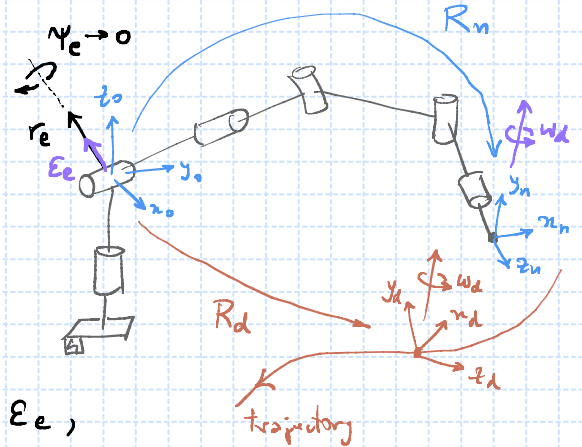
• Control action for the orientation,

$$\omega_n = \mathcal{Y}_o \dot{q}$$

Orientation error,

$$R_e R_n = R_d \Rightarrow R_e = R_d R_n^T$$

orientation error



Attribute an angular velocity meaning to $\mathcal{E}_e$,

$$\omega_d + K_o \mathcal{E}_e = \mathcal{Y}_o \dot{q},$$

$$\dot{q} = \mathcal{Y}_o^{-1} (\omega_d + K_o \mathcal{E}_e)$$

, K_o positive definite. Example $K_o = k_o \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$, $k_o > 0$.

Extraction of the quaternion's vector part,

$$\text{But } R_e = \begin{pmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{pmatrix}.$$

$$\left\{ \begin{array}{l} \eta_e = \frac{1}{2} \sqrt{|r_{11} + r_{22} + r_{33} + 1|} \\ \mathcal{E}_e = \frac{1}{2} \begin{pmatrix} \text{sign}(r_{32} - r_{23}) \sqrt{|r_{11} - r_{22} - r_{33} + 1|} \\ \text{sign}(r_{13} - r_{31}) \sqrt{|r_{22} - r_{11} - r_{33} + 1|} \\ \text{sign}(r_{21} - r_{12}) \sqrt{|r_{33} - r_{11} - r_{22} + 1|} \end{pmatrix}, \forall \eta_e \neq 0 \end{array} \right.$$

$$\text{sign}(n) = \begin{cases} 1, & n > 0 \\ 0, & n = 0 \\ -1, & n < 0 \end{cases}$$

$$|n| = \text{abs}(n)$$

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• stirring,

$$\dot{q} = \mathcal{Y}^{-1} \begin{pmatrix} \dot{p}_d + K_p(p_d - p_n) \\ \omega_d + K_o \mathcal{E}_e \end{pmatrix},$$

CLIK

$$q(t) = \int_0^t \dot{q}(u) du + q(0)$$

$$\dot{q} = \gamma^{-1} \begin{pmatrix} \dot{P}_d + K_p(P_d - P_n) \\ W_d + K_o E_e \end{pmatrix}$$

$$R_e = R_d R_n^T$$

